
Tylersoft Card Processor

Payment API Reference

Version 2.1

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Author: Gaonyadiwe Gaboiphiwe

1. Overview

The Tylersoft Card Processor (TCP) API enables card payment processing including authorization, capture, and settlement of transactions worldwide. This document is intended for application developers integrating TCP into their systems.

1.1 Base URL

HTTPS	<code>https://{domain}:{port}/tcp/api/</code>
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1.2 Onboarding Requirements

To have your organization registered in TCP, provide Tylersoft with the following:

- Organization email address
- Post-back (callback) URL

2. Authentication

All API requests require an authkey field. This key authenticates requests and ensures only authorized organizations can submit transactions.

2.1 authkey Construction

Generate the authkey using the formula below:

```
authkey = SHA512( Base64( key + tranid + amount + timestamp + phone ) )
```

2.2 Parameter Definitions

Field	Type	Description
key	String	Unique organization key — issued by Tylersoft upon onboarding.
tranid	String	Unique transaction identifier generated by your system.
amount	Decimal	Transaction amount to be debited.
timestamp	String	Transaction request time in the format YYYY-MM-DD HH:MM:SS+0000.
phone	String	Cardholder's phone number (digits only, with country code).

2.3 Example

Given the following inputs:

```
key      = !@#$%^&*()~_+}{?  
tranid   = 1919091  
amount   = 3  
timestamp = 2017-01-22 03:41:02+0000  
phone    = 254721765451
```

Concatenated string:

```
!@#$%^&*()~_+}{?191909132017-01-22 03:41:02+0000254721765451
```

Base64 of above:

```
IUAjJCVeJiooKX5fK317PzE5MTkwOTEzMjAxNy0wMS0yMiAwMzo0MTowMiswMDAwMjU0NzIxNzY1NDUx
```

SHA512 (authkey):

```
0105FC3963F29C1DCA74229D499E79CD9DB0D0FDF32B203676FD802CC9191727  
FE3166A80D95DD8893512531F37E59FB7DA853849DFA9DAAFC4AD49821C71529
```

⚠ Security: Never include the organization key (key) in any API request payload or client-side code. Exposure compromises all transactions under that key.

3. API Endpoints

TCP follows a two-phase payment flow. Both phases must be executed in order.

3.1 Device Data Collection

Collects device fingerprint information from the cardholder's browser. TCP responds with a device collection URL and access token for forwarding to the issuing bank.

POST`/tcp/api/deviceData-Collection`

Request Parameters

Field	Type	Description
tranid	String	Unique transaction ID.
country	String	Cardholder's country (e.g. Botswana).
amount	Decimal	Transaction amount.
currency	String	ISO 4217 currency code (e.g. BWP).
authkey	String	SHA512 authentication key (see Section 2).
org	String	Organization identifier provided by Tylersoft.
Serviceid	String	Service identifier provided by Tylersoft.
processingcode	String	Transaction processing code.
firstname	String	Cardholder first name.
secondname	String	Cardholder last name.
phone	String	Cardholder phone number with country code.
email	String	Cardholder email address.
card_number	String	PAN — card number (16 digits).
card_cardType	String	Card type code (e.g. 001 = Visa).
card_expiration_month	String	Card expiry month (MM).
card_expiration_year	String	Card expiry year (YYYY).
card_cvNumber	String	Card CVV/CVC number.
timestamp	String	Request timestamp (YYYY-MM-DD HH:MM:SS).

Sample Request

```
{
  "tranid": "2025-05-01 09:25:41",
  "country": "Botswana",
  "amount": "1",
  "currency": "BWP",
  "authkey": "6dc8ed39...a36645",
  "org": "tylersoft_eclectics",
  "Serviceid": "1010",
```

```
{
  "processingcode": "000010",
  "firstname": "Gao",
  "secondname": "Gabo",
  "phone": "26774374154",
  "email": "ggaboiphiwe@outlook.com",
  "card_number": "4111111111111111",
  "card_cardType": "001",
  "card_expiration_month": "12",
  "card_expiration_year": "2025",
  "card_cvNumber": "123",
  "timestamp": "2025-04-16 21:47:48"
}
```

Response Fields

Field	Type	Description
statusCode	String	00 = success.
tranid	String	Echo of the transaction ID sent in the request.
statusmessage	String	Human-readable status (e.g. COMPLETED).
id	String	Internal TCP transaction reference.
accessToken	String	JWT for the device data collection iframe POST.
deviceDataCollectionUrl	String	URL to POST the JWT from the hidden iframe.
referenceId	String	Reference ID required in the subsequent Payment request.

Sample Response

```
{
  "statusCode": "00",
  "tranid": "2025-05-01 09:25:41",
  "statusmessage": "COMPLETED",
  "id": "7464578125406450604807",
  "accessToken": "eyJhbGciOi...",
  "deviceDataCollectionUrl":
  "https://centinelapistag.cardinalcommerce.com/V1/Cruise/Collect",
  "referenceId": "ebd3166d-04ab-40b8-abbf-24712612acdc"
}
```

Client-side Device Fingerprinting

After receiving the Device Data Collection response, render a hidden iframe and POST the JWT to the deviceDataCollectionUrl:

```
<iframe name="ddc-iframe" height="1" width="1" style="display:none;"></iframe>
```

```
<form id="ddc-form" target="ddc-iframe" method="POST"
action="<<deviceDataCollectionUrl>>">
  <input type="hidden" name="JWT" value="<<accessToken>>" />
</form>
<script>
  document.getElementById('ddc-form').submit();
</script>
```

Subscribe to the window message event to confirm device collection is complete. No data from this event needs to be retained.

```
<script>
window.addEventListener('message', (event) => {
  // Test origin: https://centinelapistag.cardinalcommerce.com
  // Prod origin: https://centinelapi.cardinalcommerce.com
  if (event.origin === 'https://centinelapistag.cardinalcommerce.com') {
    let data = JSON.parse(event.data);
    if (data !== undefined && data.Status) {
      // Device profiling complete – proceed to Payment request
    }
  }
}, false);
</script>
```

3.2 Payment

Initiates payment processing. TCP authenticates the cardholder via 3D Secure and responds with a step-up URL if additional authentication is required. This endpoint must be called after Device Data Collection.

POST

/tcp/api/request

Request Parameters

Include all fields from the Device Data Collection request, plus the following additional fields:

Field	Type	Description
cbyreferenceId	String	referenceId returned by the Device Data Collection response.
ipAddress	String	Cardholder's IP address.
httpAcceptContent	String	Value of the Accept HTTP header from the cardholder's browser.
httpBrowserLanguage	String	Browser language (e.g. en_us).
httpBrowserJavaEnabled	Boolean	Whether Java is enabled in the browser.

httpBrowserJavaScriptEnabled	Boolean	Whether JavaScript is enabled.
httpBrowserColorDepth	String	Browser screen color depth in bits (e.g. 24).
httpBrowserScreenHeight	String	Screen height in pixels.
httpBrowserScreenWidth	String	Screen width in pixels.
httpBrowserTimeDifference	String	UTC offset in minutes.
userAgentBrowserValue	String	Browser User-Agent string.

Sample Request

```
{
  "tranid": "2025-05-01 09:25:41",
  "country": "Botswana",
  "amount": "1",
  "currency": "BWP",
  "cbyreferenceId": "ebd3166d-04ab-40b8-abbf-24712612acdc",
  "authkey": "6dc8ed39...a36645",
  "org": "tylersoft_eclectics",
  "Serviceid": "1010",
  "processingcode": "000010",
  "firstname": "Gao",
  "secondname": "Gabo",
  "phone": "26774374154",
  "email": "ggaboiphiwe@outlook.com",
  "card_number": "4111111111111111",
  "card_cardType": "001",
  "card_expiration_month": "12",
  "card_expiration_year": "2025",
  "card_cvNumber": "123",
  "timestamp": "2025-04-16 21:47:48",
  "ipAddress": "123.45.65.20",
  "httpAcceptContent": "text/html",
  "httpBrowserLanguage": "en_us",
  "httpBrowserJavaEnabled": "false",
  "httpBrowserJavaScriptEnabled": "true",
  "httpBrowserColorDepth": "24",
  "httpBrowserScreenHeight": "900",
  "httpBrowserScreenWidth": "1440",
  "httpBrowserTimeDifference": "120",
  "userAgentBrowserValue": "Mozilla/5.0 ..."
}
```

Response Fields

Field	Type	Description
statusCode	String	001 = authentication required. 00 = approved without step-up.

message	String	PENDING_AUTHENTICATION — cardholder must complete 3DS step-up.
pareq	String	Encoded authentication request payload for the step-up iframe.
pastepup	String	Step-up URL for the iframe form POST.
accessToken	String	JWT for the step-up iframe POST.

3DS Step-up Authentication

If the response contains message: PENDING_AUTHENTICATION, render a step-up iframe and POST to the bank's Access Control Server:

```
<iframe name="step-up-iframe" height="400" width="400"
style="display:none;"></iframe>
<form id="step-up-form" target="step-up-iframe" method="POST"
action="<<pastepup>>">
  <input type="hidden" name="JWT" value="<<accessToken>>" />
  <input type="hidden" name="MD" value="<<optional merchant data>>" />
</form>
<script>
  document.getElementById('step-up-form').submit();
</script>
```

Note: If the issuing bank does not require step-up authentication, TCP will process the transaction immediately and deliver the final status via callback — no iframe is needed.

4. Transaction Callback

The final transaction result is delivered by TCP to your registered post-back URL via HTTP POST. Your server must respond with an acknowledgement to confirm receipt.

4.1 Callback Payload (TCP → Your Server)

Field	Type	Description
statuscode	String	100 = processed successfully.
tranid	String	The original transaction ID.
status	String	Human-readable result message.

```
{
  "statuscode": "100",
```

```
"tranid": "2025-05-01 09:25:41",
"status": "Request was processed successfully"
}
```

4.2 Acknowledgement (Your Server → TCP)

Return a JSON acknowledgement confirming receipt of the callback:

```
{
  "statuscode": "00",
  "tranid": "2025-05-01 09:25:41"
}
```

5. Integration Flow

Follow these steps in order for every payment transaction:

1. Generate authkey using your organization key + transaction parameters.
2. POST to /tcp/api/deviceData-Collection and retrieve the referenceId, accessToken, and deviceDataCollectionUrl.
3. Render the hidden device fingerprint iframe and wait for the profile.completed event.
4. POST to /tcp/api/request including the cbyreferenceId and all device browser data.
5. If PENDING_AUTHENTICATION, render the step-up iframe so the cardholder can enter their OTP.
6. Receive the final callback on your post-back URL and reply with the acknowledgement payload.

6. Status Codes

Field	Type	Description
00	String	Success / Acknowledged.
001	String	Authentication required — proceed with 3DS step-up.
100	String	Transaction processed successfully (callback).

7. Environments

Field	Type	Description
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Test (Staging)	Cardinal	https://centinelapistag.cardinalcommerce.com
Production	Cardinal	https://centinelapi.cardinalcommerce.com

8. Support & Contact

Field	Type	Description
Email		enquiries@tylersoft.net
Phone		+267 395 1798
Fax		+267 395 1898
Address		Plot 75782, Pinnacle Park, Setlhoa Village, Gaborone, Botswana
PO Box		Private Bag BO 70, Gaborone, Botswana
